

Technical Project Report: Automated Real-Time Mini-Security Triage Station

By: Lakunsin

Entry-Level Cybersecurity Analyst

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Executive Summary

This report documents the design, implementation, and refinement of an automated Mini-Security Triage Station deployed within a Linux environment (Ubuntu VM). The system provides Security Operations Center (SOC) style real-time analysis by bridging local OS telemetry with the VirusTotal v3 API. By combining automated clipboard monitoring (xclip + Regex optimization) and real-time filesystem observation (watchdog), the triage station instantly checks copied URLs/domains and newly downloaded files against global threat intelligence databases, utilizing desktop notifications (notify-send) and an automated quarantine/deletion policy for verified malicious files.

System Architecture

This triage station relies on a combination of native Linux utilities, configuration files, and Python scripts working together:

- **Clipboard Utility (xclip):** A lightweight background Linux tool that allows the system to read clipboard memory. It spies on copied text so that links or domains can be automatically processed.
- **Desktop Notification Tool (notify-send):** A native Linux utility that generates instant desktop pop-up alerts. This is used to immediately inform the user of VirusTotal scan results without needing the terminal open.
- **Secure Configuration File (.env):** A hidden file used to store the private VirusTotal API key. This keeps sensitive credentials secure and separate from the source code.
- **Python Dependencies:**
 - **pyperclip:** Allows Python to interact directly with the clipboard.
 - **requests:** Used to send and retrieve data from the VirusTotal API.
 - **python-dotenv:** Enables Python to securely read credentials from the .env file.
 - **watchdog:** Monitors the target directory for filesystem changes.
- **The Scanning Core (scanner.py):** The engine of the project. It handles the communication with VirusTotal, analyses URLs and file hashes, coordinates desktop notifications, and executes the auto-deletion logic.
- **The Clipboard Watcher (clipboard_watcher.py):** A background script that continually checks the clipboard, uses a regular expression to extract URLs or naked domains and forwards them to the scanner.
- **The Directory Observer (downloads_monitor.py):** A background script that watches the local *~/Downloads folder*. The moment a new file is fully written to the disk, it triggers the scanning core to evaluate the file's hash.

Phase 1: Automated Clipboard Monitoring with VirusTotal API

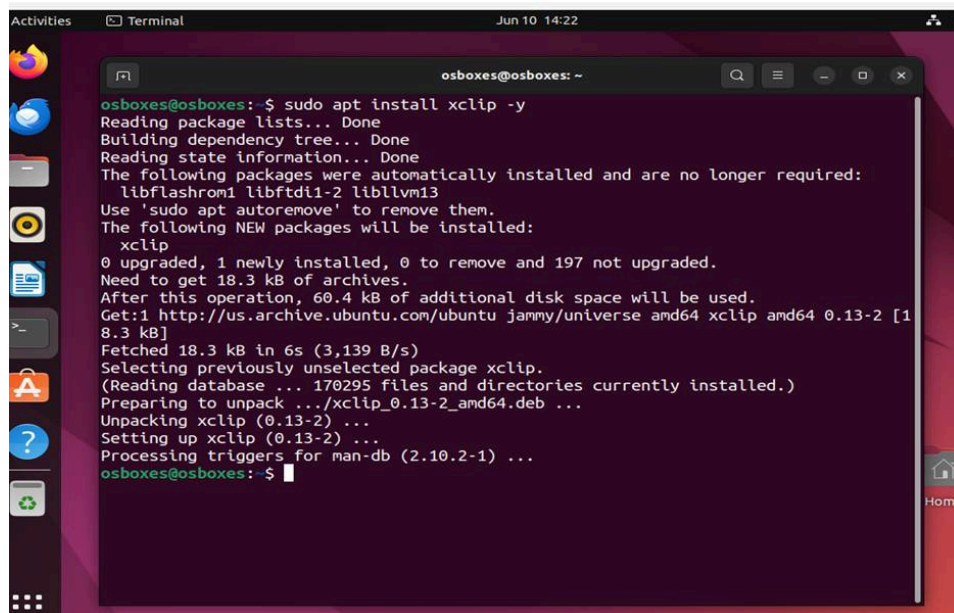
Step 1: Installing the Linux Clipboard Utility (xclip)

To allow the background process to monitor clipboard memory and capture copied URL/links, install the lightweight clipboard manager xclip

Open the terminal and run this command to install it:

```
sudo apt update
```

```
sudo apt install xclip -y
```



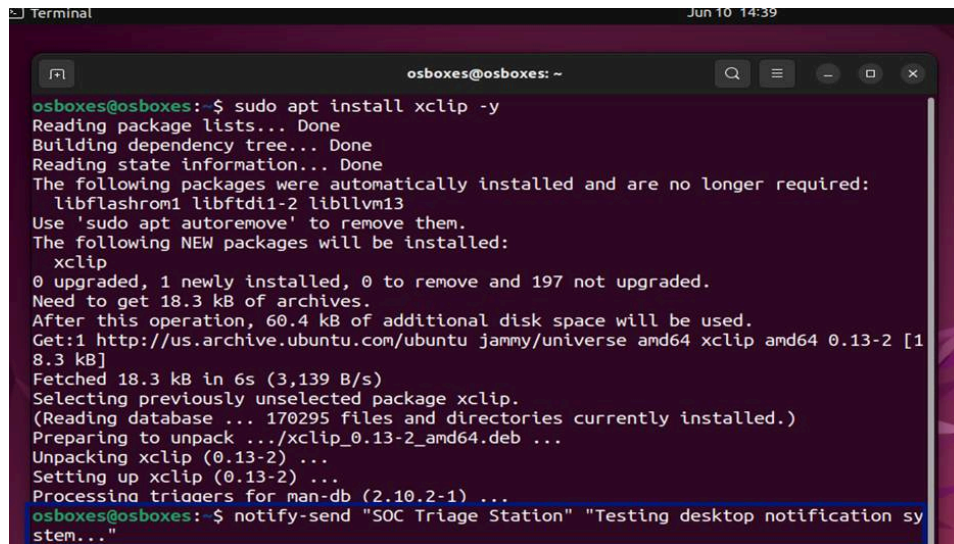
```
osboxes@osboxes: ~  
osboxes@osboxes:~$ sudo apt install xclip -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following packages were automatically installed and are no longer required:  
  libflashrom1 libftdi1-2 libllvm13  
Use 'sudo apt autoremove' to remove them.  
The following NEW packages will be installed:  
  xclip  
0 upgraded, 1 newly installed, 0 to remove and 197 not upgraded.  
Need to get 18.3 kB of archives.  
After this operation, 60.4 kB of additional disk space will be used.  
Get:1 http://us.archive.ubuntu.com/ubuntu jammy/universe amd64 xclip amd64 0.13-2 [18.3 kB]  
Fetched 18.3 kB in 6s (3,139 B/s)  
Selecting previously unselected package xclip.  
(Reading database ... 170295 files and directories currently installed.)  
Preparing to unpack .../xclip_0.13-2_amd64.deb ...  
Unpacking xclip (0.13-2) ...  
Setting up xclip (0.13-2) ...  
Processing triggers for man-db (2.10.2-1) ...  
osboxes@osboxes:~$
```

Step 2: Installing the Desktop Notification Tool (notify-send)

To display scan alerts on your screen in real time without relying on an open terminal window, verify that your system is equipped with notify-send

Run this command:

```
notify-send "SOC Triage Station" "Testing desktop notification system..."
```



```
Terminal Jun 10 14:39
osboxes@osboxes: ~
osboxes@osboxes:~$ sudo apt install xclip -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libflashrom1 libftdi1-2 liblvm13
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  xclip
0 upgraded, 1 newly installed, 0 to remove and 197 not upgraded.
Need to get 18.3 kB of archives.
After this operation, 60.4 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu jammy/universe amd64 xclip amd64 0.13-2 [18.3 kB]
Fetched 18.3 kB in 6s (3,139 B/s)
Selecting previously unselected package xclip.
(Reading database ... 170295 files and directories currently installed.)
Preparing to unpack .../xclip_0.13-2_amd64.deb ...
Unpacking xclip (0.13-2) ...
Setting up xclip (0.13-2) ...
Processing triggers for man-db (2.10.2-1) ...
osboxes@osboxes:~$ notify-send "SOC Triage Station" "Testing desktop notification system..."
```

Step 3: Create a Fresh Project Directory

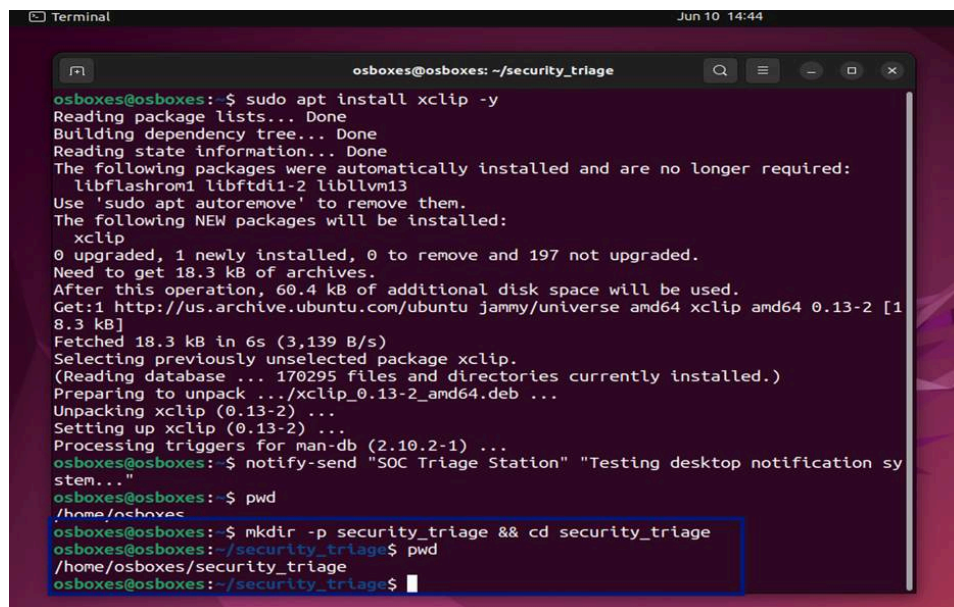
To keep the triage system organized, a dedicated project directory is created:

Run this command to create the folder and move inside it:

```
mkdir -p ~/security_triage
```

```
cd ~/security_triage
```

(Verify that your terminal prompt changes to show ~/security_triage before moving forward.)



```
Terminal Jun 10 14:44
osboxes@osboxes: ~/security_triage
osboxes@osboxes:~$ sudo apt install xclip -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libflashrom1 libftdi1-2 liblvm13
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  xclip
0 upgraded, 1 newly installed, 0 to remove and 197 not upgraded.
Need to get 18.3 kB of archives.
After this operation, 60.4 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu jammy/universe amd64 xclip amd64 0.13-2 [18.3 kB]
Fetched 18.3 kB in 6s (3,139 B/s)
Selecting previously unselected package xclip.
(Reading database ... 170295 files and directories currently installed.)
Preparing to unpack .../xclip_0.13-2_amd64.deb ...
Unpacking xclip (0.13-2) ...
Setting up xclip (0.13-2) ...
Processing triggers for man-db (2.10.2-1) ...
osboxes@osboxes:~$ notify-send "SOC Triage Station" "Testing desktop notification system..."
osboxes@osboxes:~$ pwd
/home/osboxes
osboxes@osboxes:~$ mkdir -p security_triage && cd security_triage
osboxes@osboxes:~/security_triage$ pwd
/home/osboxes/security_triage
osboxes@osboxes:~/security_triage$
```

Step 4: Create the Secure Configuration File (.env)

To query VirusTotal safely, a private API key is required. To protect this key from exposure in your scripts, save it as an environment variable in a hidden .env file

Run this command:

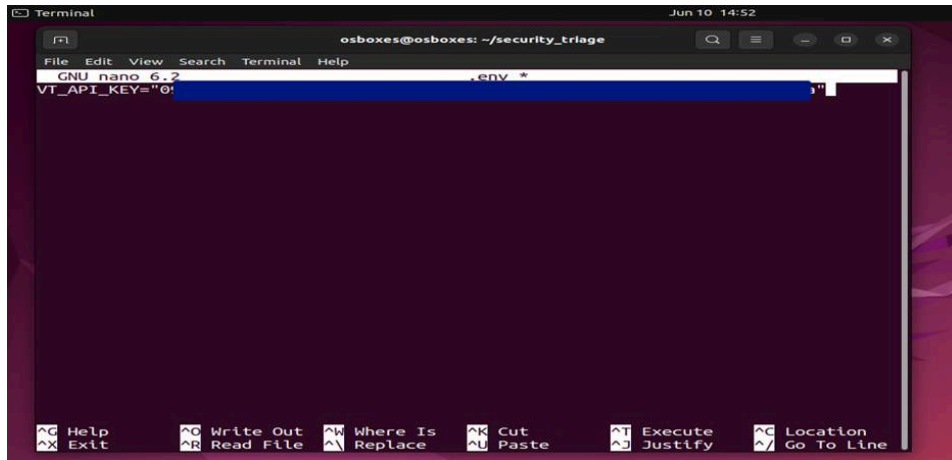
nano .env

Inside the nano editor, paste your API key matching this format

VT_API_KEY=YOUR_ACTUAL_API_KEY_HERE

Save (Press Ctrl + O, then press Enter)

Exit (Press Ctrl + X)



Step 5: Install Python Dependencies

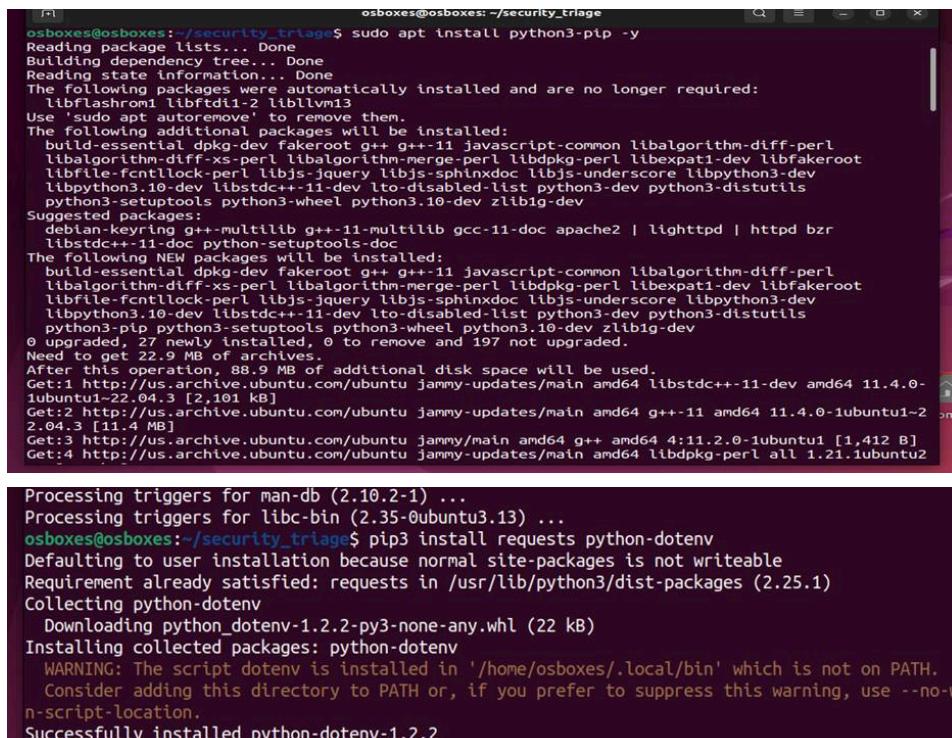
Before running Python packages, install pip3 (the Python package installer), followed by the specific libraries needed to securely connect to the VirusTotal API

Run this:

```
sudo apt install python3-pip -y
```

Then install the Python packages:

```
pip3 install requests python-dotenv
```



```
osboxes@osboxes:~/security_triage$ sudo apt install python3-pip -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libflashrom1 libftdi1-2 liblvm13
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  build-essential dpkg-dev fakeroot g++ g++-11 javascript-common libalgorithm-diff-perl
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libdpkg-perl libexpat1-dev libfakeroot
  libfile-fcntllock-perl libjs-jquery libjs-sphinxdoc libjs-underscore libpython3-dev
  libpython3.10-dev libstdc++-11-dev lto-disabled-list python3-dev python3-distutils
  python3-setuptools python3-wheel python3.10-dev zlib1g-dev
Suggested packages:
  debian-keyring g++-multilib g++-11-multilib gcc-11-doc apache2 | lighttpd | httpd bzip
  libstdc++-11-doc python-setuptools-doc
The following NEW packages will be installed:
  build-essential dpkg-dev fakeroot g++ g++-11 javascript-common libalgorithm-diff-perl
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libdpkg-perl libexpat1-dev libfakeroot
  libfile-fcntllock-perl libjs-jquery libjs-sphinxdoc libjs-underscore libpython3-dev
  libpython3.10-dev libstdc++-11-dev lto-disabled-list python3-dev python3-distutils
  python3-pip python3-setuptools python3-wheel python3.10-dev zlib1g-dev
0 upgraded, 27 newly installed, 0 to remove and 197 not upgraded.
Need to get 22.9 MB of archives.
After this operation, 88.9 MB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libstdc++-11-dev amd64 11.4.0-
1ubuntu1-22.04.3 [2,101 kB]
Get:2 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 g++-11 amd64 11.4.0-1ubuntu1-2
2.04.3 [11.4 MB]
Get:3 http://us.archive.ubuntu.com/ubuntu jammy/main amd64 g++ amd64 4:11.2.0-1ubuntu1 [1,412 B]
Get:4 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libdpkg-perl all 1.21.1ubuntu2
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.13) ...
osboxes@osboxes:~/security_triage$ pip3 install requests python-dotenv
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: requests in /usr/lib/python3/dist-packages (2.25.1)
Collecting python-dotenv
  Downloading python_dotenv-1.2.2-py3-none-any.whl (22 kB)
Installing collected packages: python-dotenv
  WARNING: The script dotenv is installed in '/home/osboxes/.local/bin' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-w
  n-script-location.
Successfully installed python-dotenv-1.2.2
```

Step 6: Create the Core Scanner Script (scanner.py)

This Python script is designed to handle two critical tasks:

1. Analyze a Link
2. Analyze a file hash

It reads the .env file for your private API key, queries VirusTotal, and sends a "Safe", "Suspicious", or "Malicious" pop-up result directly to your desktop

Run this command in the terminal:

```
nano scanner.py
```

Copy the entire block of code below and paste it directly into the nano editor:

```
import os
import sys
import base64
import hashlib
import requests
import subprocess

from dotenv import load_dotenv

# Load API key
load_dotenv()

API_KEY = os.getenv("VT_API_KEY")

def notify(title, message):
    subprocess.run(["notify-send", title, message])

def check_vt(endpoint, target_id, label):
    if not API_KEY:
        notify("VT Error", "Missing API key in .env")
        return

    headers = {
        "x-apikey": API_KEY
    }

    url = f"https://www.virustotal.com/api/v3/{endpoint}/{target_id}"

    try:
        response = requests.get(url, headers=headers)

        if response.status_code == 200:
            data = response.json()
            stats = data["data"]["attributes"]["last_analysis_stats"]
            malicious = stats.get("malicious", 0)
            suspicious = stats.get("suspicious", 0)
            harmless = stats.get("harmless", 0)

            if malicious > 0:
                verdict = "MALICIOUS"
                msg = f"{label}: {malicious} engines flagged it"

            elif suspicious > 0:
                verdict = "SUSPICIOUS"
                msg = f"{label}: {suspicious} engines suspicious"

            else:
                verdict = "SAFE"
                msg = f"{label}: Clean ({harmless} engines)"
```

```

        notify(f"VT RESULT: {verdict}", msg)

    elif response.status_code == 404:

        notify("VT RESULT", f"{label}: Not found in database")

    else:

        notify("VT ERROR", f"Status code {response.status_code}")

except Exception as e:

    notify("VT ERROR", str(e))

def scan_url(url):

    url_id = base64.urlsafe_b64encode(url.encode()).decode().strip("=")

    check_vt("urls", url_id, "URL")

def scan_file(path):

    if not os.path.exists(path):

        notify("ERROR", "File not found")

        return

    sha256 = hashlib.sha256()

    with open(path, "rb") as f:

        for chunk in iter(lambda: f.read(4096), b""):

            sha256.update(chunk)

    file_hash = sha256.hexdigest()

    check_vt("files", file_hash, "FILE")

if name == "main":

    if len(sys.argv) < 3:

        print("Usage: python3 scanner.py --url OR --file")

        sys.exit(1)

    mode = sys.argv[1]

    target = sys.argv[2]

    if mode == "--url":

        scan_url(target)

    elif mode == "--file":

        scan_file(target)

```

Save (Ctrl + O, Enter)

Exit (Ctrl + X)

```

osboxes@osboxes: ~/security_triage
GNU nano 6.2 scanner.py *
import os
import sys
import base64
import hashlib
import requests
import subprocess
from dotenv import load_dotenv

# Load API key
load_dotenv()
API_KEY = os.getenv("VT_API_KEY")

def notify(title, message):
    subprocess.run(["notify-send", title, message])

def check_vt(endpoint, target_id, label):
    if not API_KEY:
        notify("VT Error", "Missing API key in .env")
        return

    headers = {
        "x-apikey": API_KEY
    }

    url = f"https://www.virustotal.com/api/v3/{endpoint}/{target_id}"

```

Step 7: Test URL Scanning Manually

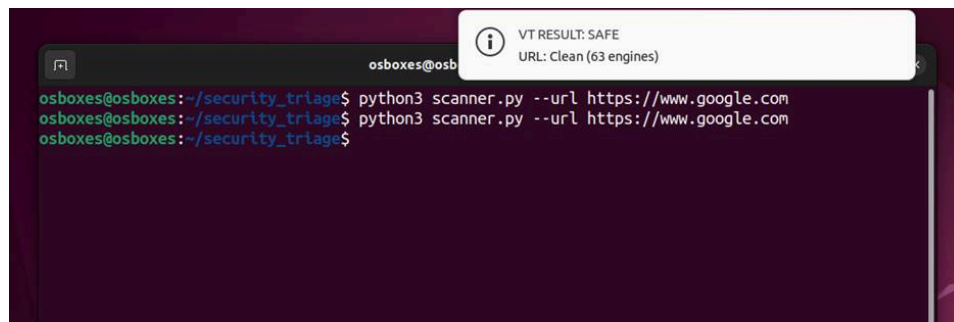
Before running the background monitor, verify that scanner.py can manually reach VirusTotal and generate a desktop alert for a URL.

Run this command in your terminal:

```
python3 scanner.py --url https://www.google.com
```

What to look for:

- Check your desktop for a notification from **VT RESULT: SAFE**.



Step 8 — Install clipboard watcher dependency

Run this:

```
sudo apt install xclip xsel -y
```

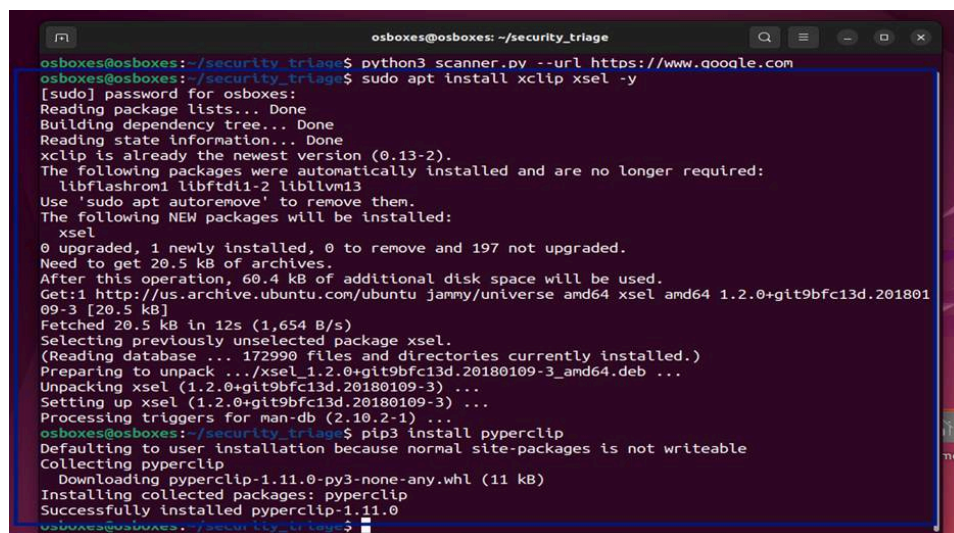
Then install Python clipboard tool:

```
pip3 install pyperclip
```

Why this matters

- pyperclip = reads anything you copy (Ctrl+C)

This is what turns the script into a real-time SOC triage agent



Step 9 - Create Clipboard Monitor (Auto VirusTotal Scanner)

This script will:

- Watch everything you copy
- Detect URLs automatically
- Send them to your existing scanner.py
- Show instant SOC-style popups

Create the watcher file

Run:

nano clipboard_watcher.py

Paste this code

```
import time

import subprocess

import pyperclip

import re

last_clipboard = ""

# simple URL detection pattern
url_pattern = re.compile(r"https?://[^\s]+")

print("Clipboard watcher running... (Ctrl+C to stop)")

while True:

    try:

        clip = pyperclip.paste()

        if clip != last_clipboard:

            last_clipboard = clip

            match = url_pattern.search(clip)

            if match:

                url = match.group(0)

                print(f"[+] URL detected: {url}")

                subprocess.run(["python3", "scanner.py", "--url", url])

            time.sleep(2)

    except KeyboardInterrupt:

        print("Stopped.")

        break

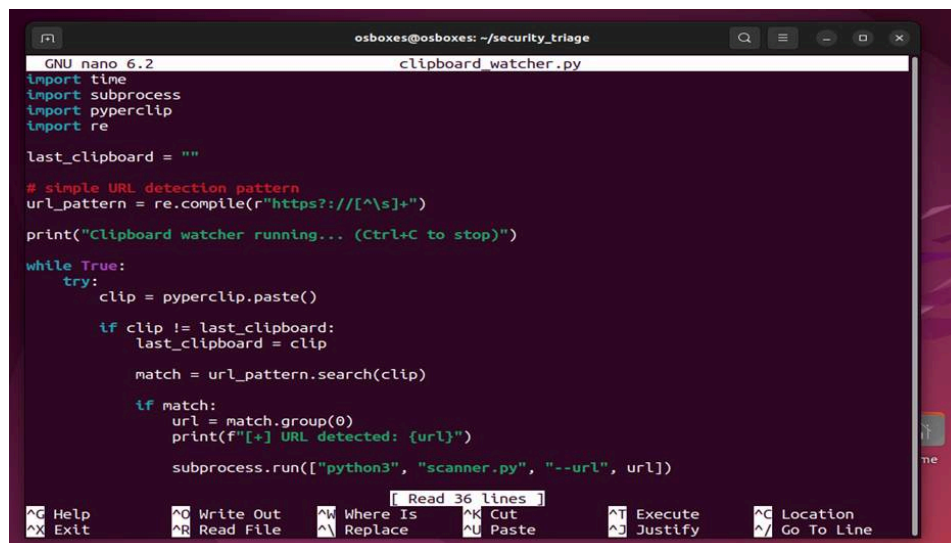
    except Exception as e:

        print("Error:", e)

        time.sleep(2)
```

Save: Ctrl + O, Enter

Exit: Ctrl + X



```
osboxes@osboxes: ~/security_triage
GNU nano 6.2 clipboard_watcher.py
import time
import subprocess
import pyperclip
import re

last_clipboard = ""

# simple URL detection pattern
url_pattern = re.compile(r"https?://[^\s]+")

print("Clipboard watcher running... (Ctrl+C to stop)")

while True:
    try:
        clip = pyperclip.paste()

        if clip != last_clipboard:
            last_clipboard = clip

            match = url_pattern.search(clip)

            if match:
                url = match.group(0)
                print(f"[+] URL detected: {url}")
                subprocess.run(["python3", "scanner.py", "--url", url])

            time.sleep(2)

    except KeyboardInterrupt:
        print("Stopped.")
        break

    except Exception as e:
        print("Error:", e)
        time.sleep(2)
```

me

Step 10 — Run your SOC real-time engine

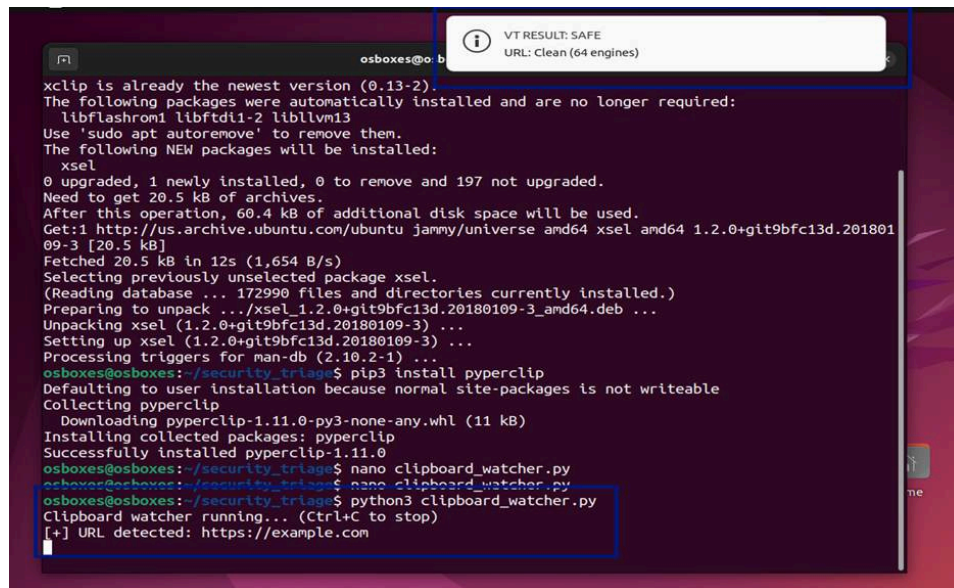
Start it in the terminal:

```
python3 clipboard_watcher.py
```

How to test it

1. Copy any link like: <https://example.com>
2. You should see VirusTotal popup result in the VM

Press Ctrl C to stop watcher



```
osboxes@osboxes:~$ xclip is already the newest version (0.13-2).
The following packages were automatically installed and are no longer required:
  libflashrom1 libftdi1-2 liblvm13
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  xsel
0 upgraded, 1 newly installed, 0 to remove and 197 not upgraded.
Need to get 20.5 kB of archives.
After this operation, 60.4 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu jammy/universe amd64 xsel amd64 1.2.0+git9bfc13d.20180109-3 [20.5 kB]
Fetched 20.5 kB in 12s (1,654 B/s)
Selecting previously unselected package xsel.
(Reading database ... 172990 files and directories currently installed.)
Preparing to unpack .../xsel_1.2.0+git9bfc13d.20180109-3_amd64.deb ...
Unpacking xsel (1.2.0+git9bfc13d.20180109-3) ...
Setting up xsel (1.2.0+git9bfc13d.20180109-3) ...
Processing triggers for man-db (2.10.2-1) ...
osboxes@osboxes:~/security_triage$ pip3 install pyperclip
Defaulting to user installation because normal site-packages is not writeable
Collecting pyperclip
  Downloading pyperclip-1.11.0-py3-none-any.whl (11 kB)
Installing collected packages: pyperclip
Successfully installed pyperclip-1.11.0
osboxes@osboxes:~/security_triage$ nano clipboard_watcher.py
osboxes@osboxes:~/security_triage$ nano clipboard_watcher.py
osboxes@osboxes:~/security_triage$ python3 clipboard_watcher.py
Clipboard watcher running... (Ctrl+C to stop)
[+] URL detected: https://example.com
```

Fixing the Detection Pattern & Enhancing Alerts

I noticed this only works for links with https:// or http:// and the popup message didn't display the link.

A. Fix the detection pattern

Make the system smarter so it catches naked domains such as:

- <https://example.com>
- paypa1.com

Open the watcher:

```
nano clipboard_watcher.py
```

Find this line:

```
url_pattern = re.compile(r"https?://\s+")
```

Replace it with this:

```
url_pattern = re.compile(r"(https?://\s+|www.\s+|[a-zA-Z0-9.-]+\.[a-zA-Z]{2,})")
```

Save: Ctrl + O, Enter

Exit: Ctrl + X


```
osboxes@osboxes: ~/security_triage
GNU nano 6.2 clipboard_watcher.py *
import time
import subprocess
import pyperclip
import re

last_clipboard = ""

# simple URL detection pattern
url_pattern = re.compile(r"(https?://[^\s]+|www\.[^\s]+[a-zA-Z0-9.-]+\.[a-zA-Z]{2,})")

print("Clipboard watcher running... (Ctrl+C to stop)")

while True:
    try:
        clip = pyperclip.paste()

        if clip != last_clipboard:
            last_clipboard = clip

            match = url_pattern.search(clip)

            if match:
                url = match.group(0)
                print(f"[+] URL detected: {url}")

                subprocess.run(["python3", "scanner.py", "--url", url])

            time.sleep(2)

    except KeyboardInterrupt:
        print("Stopped.")
        break

    except Exception as e:
        print("Error:", e)
        time.sleep(2)
```

B. Notification Context:

Modifying the desktop alert to include the tested domain/link inside the notification body.

Open the scanner file:

nano scanner.py

i. Under `def check_vt`

Change:

`notify(f"VT RESULT: {verdict}", msg)`

To:

`notify(f"VT RESULT: {verdict}", f"{msg}\n\nLINK: {label}")`

```
osboxes@osboxes: ~/security_triage
GNU nano 6.2 scanner.py
def check_vt(endpoint, target_id, label):
    if not API_KEY:
        notify("VT Error", "Missing API key in .env")
        return

    headers = {
        "x-apikey": API_KEY
    }

    url = f"https://www.virustotal.com/api/v3/{endpoint}/{target_id}"

    try:
        response = requests.get(url, headers=headers)

        if response.status_code == 200:
            data = response.json()
            stats = data["data"]["attributes"]["last_analysis_stats"]

            malicious = stats.get("malicious", 0)
            suspicious = stats.get("suspicious", 0)
            harmless = stats.get("harmless", 0)

            if malicious > 0:
                verdict = "MALICIOUS"
                msg = f"{label}: {malicious} engines flagged it"
            elif suspicious > 0:
                verdict = "SUSPICIOUS"
                msg = f"{label}: {suspicious} engines suspicious"
            else:
                verdict = "SAFE"
                msg = f"{label}: Clean ({harmless} engines)"

            notify(f"VT RESULT: {verdict}", f"{msg}\n\nLINK: {label}")

        elif response.status_code == 404:
            notify("VT RESULT", f"{label}: Not found in database")
        else:
            pass

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^D Justify    ^_ Go To Line
```

ii. Still under `def check_vt`

Change:

```
def check_vt(endpoint, target_id, label):
```

To:

```
def check_vt(endpoint, target_id, label, original_value=None):
```

```
osboxes@osboxes: ~/security_triage
GNU nano 6.2 scanner.py
import os
import sys
import base64
import hashlib
import requests
import subprocess
from dotenv import load_dotenv

# Load API key
load_dotenv()
API_KEY = os.getenv("VT_API_KEY")

def notify(title, message):
    subprocess.run(["notify-send", title, message])

def check_vt(endpoint, target_id, label, original_value=None):
    if not API_KEY:
        notify("VT Error", "Missing API key in .env")
        return

    headers = {
        "x-apikey": API_KEY
    }

    url = f"https://www.virustotal.com/api/v3/{endpoint}/{target_id}"

    try:
        response = requests.get(url, headers=headers)

        if response.status_code == 200:
            data = response.json()
            stats = data["data"]["attributes"]["last_analysis_stats"]

            malicious = stats.get("malicious", 0)
            suspicious = stats.get("suspicious", 0)
            harmless = stats.get("harmless", 0)

            if malicious > 0:
                verdict = "MALICIOUS"
                msg = f"{label}: {malicious} engines flagged it"
            elif suspicious > 0:
                verdict = "SUSPICIOUS"
                msg = f"{label}: {suspicious} engines suspicious"
            else:
                verdict = "SAFE"
                msg = f"{label}: Clean ({harmless} engines)"

            notify(f"VT RESULT: {verdict}", f"{msg}\n\nLINK: {label}")

        elif response.status_code == 404:
            notify("VT RESULT", f"{label}: Not found in database")
        else:
            pass

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^D Justify    ^_ Go To Line
```

iii. URL part

Under `def scan URL`

Change:

```
check_vt("urls", url_id, "URL")
```

To:

```
check_vt("urls", url_id, "URL", url)
```

```
osboxes@osboxes: ~/security_triage
GNU nano 6.2 scanner.py *
stats = data["data"]["attributes"]["last_analysis_stats"]

malicious = stats.get("malicious", 0)
suspicious = stats.get("suspicious", 0)
harmless = stats.get("harmless", 0)

if malicious > 0:
    verdict = "MALICIOUS"
    msg = f"{label}: {malicious} engines flagged it"
elif suspicious > 0:
    verdict = "SUSPICIOUS"
    msg = f"{label}: {suspicious} engines suspicious"
else:
    verdict = "SAFE"
    msg = f"{label}: Clean ({harmless} engines)"

notify(f"VT RESULT: {verdict}", f"{msg}\n\nLINK: {label}")

elif response.status_code == 404:
    notify("VT RESULT", f"{label}: Not found in database")
else:
    notify("VT ERROR", f"Status code {response.status_code}")

except Exception as e:
    notify("VT ERROR", str(e))

def scan_url(url):
    url_id = base64.urlsafe_b64encode(url.encode()).decode().strip("=")
    check_vt("urls", url_id, "URL", url)

def scan_file(path):
    if not os.path.exists(path):
        notify("ERROR", "File not found")
        return

    sha256 = hashlib.sha256()
```

iv. Update notify line

Change:

```
notify(f"VT RESULT: {verdict}", f"{msg}\n\nLINK: {label}")
```

To:

```
notify(f"VT RESULT: {verdict}", f"{msg}\n\nLINK: {original_value or label}")
```

Save: Ctrl + O, Enter

Exit: Ctrl + X

```
osboxes@osboxes: ~/security_triage
GNU nano 6.2 scanner.py *
    notify("VT Error", "Missing API key in .env")
    return

headers = {
    "x-apikey": API_KEY
}

url = f"https://www.virustotal.com/api/v3/{endpoint}/{target_id}"

try:
    response = requests.get(url, headers=headers)

    if response.status_code == 200:
        data = response.json()
        stats = data["data"]["attributes"]["last_analysis_stats"]

        malicious = stats.get("malicious", 0)
        suspicious = stats.get("suspicious", 0)
        harmless = stats.get("harmless", 0)

        if malicious > 0:
            verdict = "MALICIOUS"
            msg = f"{label}: {malicious} engines flagged it"
        elif suspicious > 0:
            verdict = "SUSPICIOUS"
            msg = f"{label}: {suspicious} engines suspicious"
        else:
            verdict = "SAFE"
            msg = f"{label}: Clean ({harmless} engines)"

        notify(f"VT RESULT: {verdict}", f"{msg}\n\nLINK: {original_value or label}")

    elif response.status_code == 404:
        notify("VT RESULT", f"{label}: Not found in database")
    else:
        notify("VT ERROR", f"Status code {response.status_code}")

except Exception as e:
```

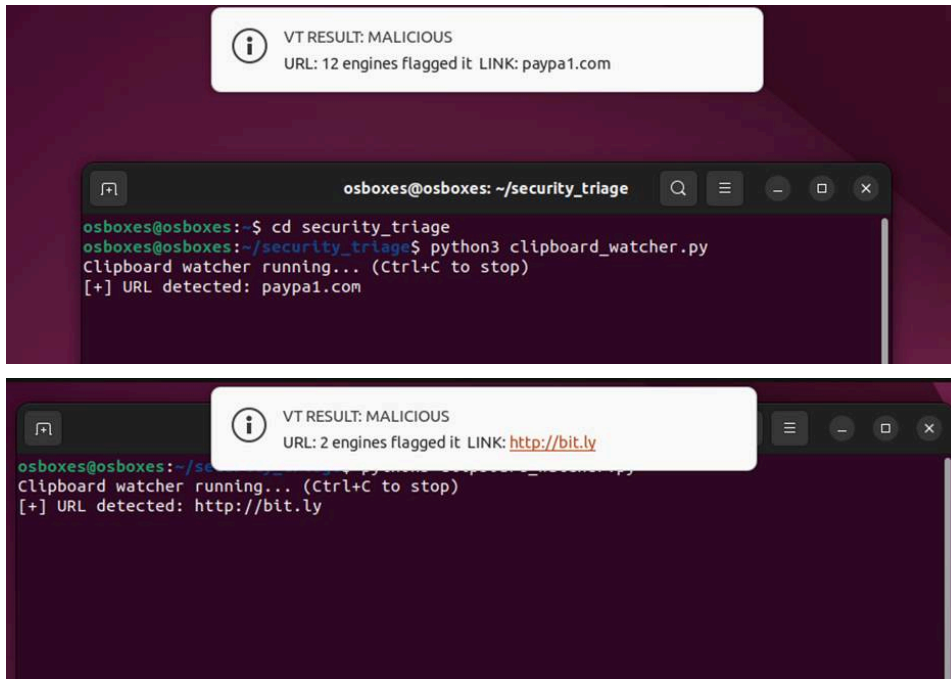
Test the fixes

Run the watcher:

python3 clipboard_watcher.py

Copy a naked domain and verify the link shows in the popup message.

Press Ctrl C to stop watcher.



Phase 2: File Download Monitoring and Malware Containment

Whenever a file hits the Downloads folder, the station will hash it, scan it via Virustotal, and automatically delete it if it is flagged as malicious by 3 or more vendors.

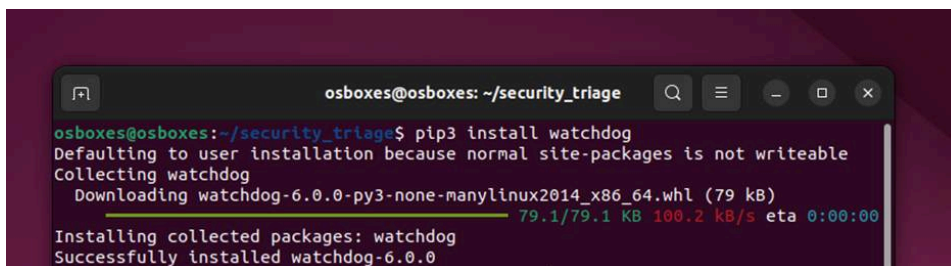
Step 11 — Install file monitoring tool

Use watchdog. This allows Python to:

- watch the Downloads folder in real time
- detect new files instantly
- trigger Virustotal scanner automatically

Run this:

pip3 install watchdog



Confirm the Downloads folder exists before the next step

ls ~/Downloads

Step 12: Create File Download Monitor (downloads_monitor.py)

Now create the file monitoring script that uses watchdog to continuously observe your ~/Downloads directory for any new files.

Run this command:

nano downloads_monitor.py

Paste this code into downloads_monitor.py

```
from watchdog.observers import Observer
from watchdog.events import FileSystemEventHandler
import subprocess
import time
import os

DOWNLOADS = os.path.expanduser("~/Downloads")

class DownloadHandler(FileSystemEventHandler):

    def on_created(self, event):

        if event.is_directory:

            return

        print(f"[+] New file detected: {event.src_path}")

        subprocess.run([

            "python3",

            "scanner.py",

            "--file",

            event.src_path

        ])

observer = Observer()

observer.schedule(DownloadHandler(), DOWNLOADS, recursive=False)

print(f"Monitoring: {DOWNLOADS}")

observer.start()

try:

    while True:

        time.sleep(1)

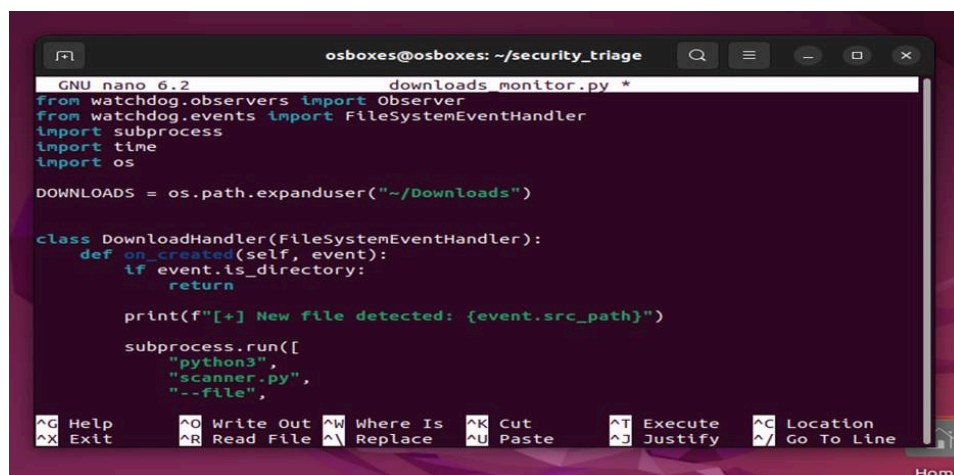
except KeyboardInterrupt:

    observer.stop()

observer.join()
```

Save: Ctrl + O, Enter

Exit: Ctrl + X



Step 14: Start the Downloads Monitor

Start the background watcher script to observe your ~/Downloads directory for any newly dropped files.

Run:

```
python3 downloads_monitor.py
```

You should see the startup log:

Monitoring: /home/osboxes/Downloads meaning the monitor is running and watching

Create a test file in Downloads

Open a second terminal (leave the first terminal running).

In the second terminal, run:

```
touch ~/Downloads/test_download.txt
```

You should get this in the first terminal:

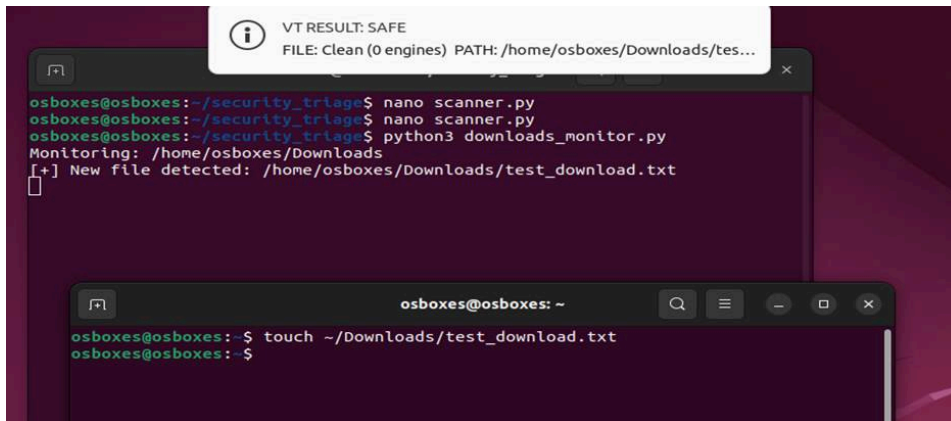
[+] New file detected: /home/osboxes/Downloads/test_download.txt

And a popup message showing:

VT RESULT: SAFE

FILE: clean (0 engines) PATH: /home/osboxes/Downloads/test_download.txt

Ctrl C to stop



Step 15 — Add SAFE delete function

Add a dedicated containment function that safely handles automated file removal when a threat is identified.

a. Open the scanner:

```
nano scanner.py
```

Add this function between `def notify()` and `def check_vt()`

```
def safe_delete_file(file_path, malicious_count):  
    print(f"[DEBUG] delete called with: {file_path} | count={malicious_count}")  
  
    if malicious_count is None or malicious_count < 3:  
        return  
  
    if file_path and os.path.exists(file_path):  
        try:  
            os.remove(file_path)  
  
            # Extract just the clean file name for the deletion alert popup
```

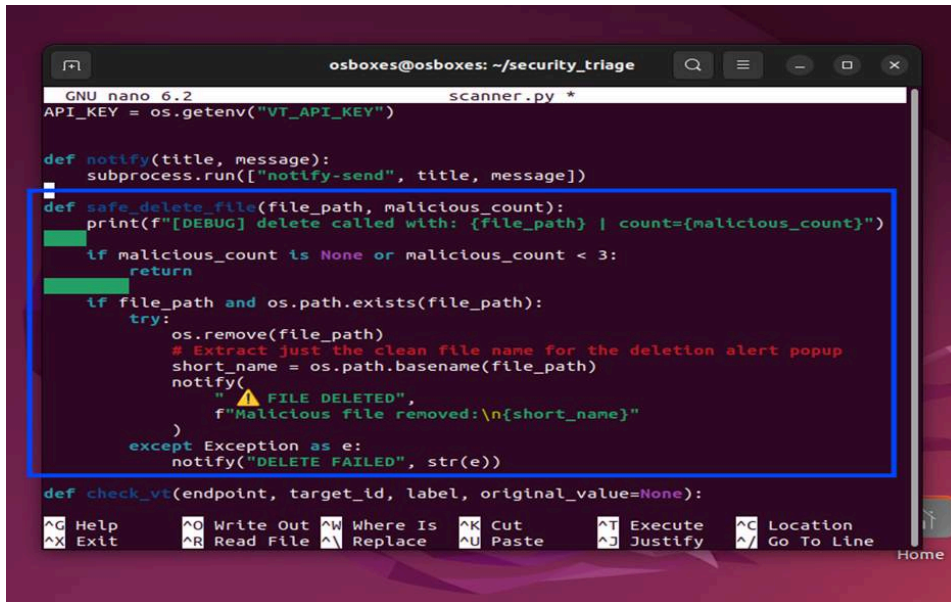
```

short_name = os.path.basename(file_path)

notify(
    " ⚠️ FILE DELETED",
    f"Malicious file removed:\n{short_name}"
)

except Exception as e:
    notify("DELETE FAILED", str(e))

```



b. Update the `check_vt` function logic

To trigger `safe_delete_file` whenever VirusTotal returns 3 or more malicious flags for a scanned file. Inside the `check_vt` function, find this exact block of code:

if malicious > 0:

```

    verdict = "MALICIOUS"

    msg = f"{label}: {malicious} engines flagged it"

```

Replace with

if malicious >= 3:

```

    verdict = "MALICIOUS"

    # Format clean file name for the popup if it's a file path
    if label == "FILE" and original_value:
        display_name = os.path.basename(original_value)
    else:
        display_name = original_value or target_id

    msg = f"{display_name}: {malicious} engines flagged it (AUTO-DELETE)"

    # Pass the full path to the background deletion logic
    safe_delete_file(original_value, malicious)

elif malicious > 0:
    verdict = "MALICIOUS"

    msg = f"{label}: {malicious} engines flagged it"

```

Save: Ctrl + O, Enter

Exit: Ctrl + X

```
osboxes@osboxes: ~/security_triage
GNU nano 6.2 scanner.py *
data = response.json()
stats = data["data"]["attributes"]["last_analysis_stats"]

malicious = stats.get("malicious", 0)
suspicious = stats.get("suspicious", 0)
harmless = stats.get("harmless", 0)

if malicious >= 3:
    verdict = "MALICIOUS"
    # Format clean file name for the popup if it's a file path
    if label == "FILE" and original_value:
        display_name = os.path.basename(original_value)
    else:
        display_name = original_value or target_id
    msg = f"{display_name}: {malicious} engines flagged it (AUTO-DELETE)"
    # Pass the full path to the background deletion logic
    safe_delete_file(original_value, malicious)
elif malicious > 0:
    verdict = "MALICIOUS"
    msg = f"{label}: {malicious} engines flagged it"
elif suspicious > 0:
    verdict = "SUSPICIOUS"
    msg = f"{label}: {suspicious} engines suspicious"
else:
    verdict = "SAFE"
    msg = f"{label}: Clean ({harmless} engines)"

notify(f"VT RESULT: {verdict}", f"{msg}\n\nLINK: {original_value or label}")

elif response.status_code == 404:
    notify("VT RESULT", f"{label}: Not found in database")
else:
    notify("VT ERROR", f"Status code {response.status_code}")

except Exception as e:
    notify("VT ERROR", str(e))

^C Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
Show Applications ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

Test

Put the entire automated pipeline to the test using the standard, harmless EICAR anti-malware test file. This simulates downloading a malicious file to verify that your script detects, hashes, queries, and automatically deletes the threat without any manual intervention.

Start the download monitor script

```
python3 downloads_monitor.py
```

Use firefox browser or open a second terminal, Download **EICAR.COM.ZIP** from

<https://www.eicar.org/download-anti-malware-testfile/>

The terminal will show:

```
[DEBUG] delete called with: /home/osboxes/Downloads/eicar_com.zip | count=60
```

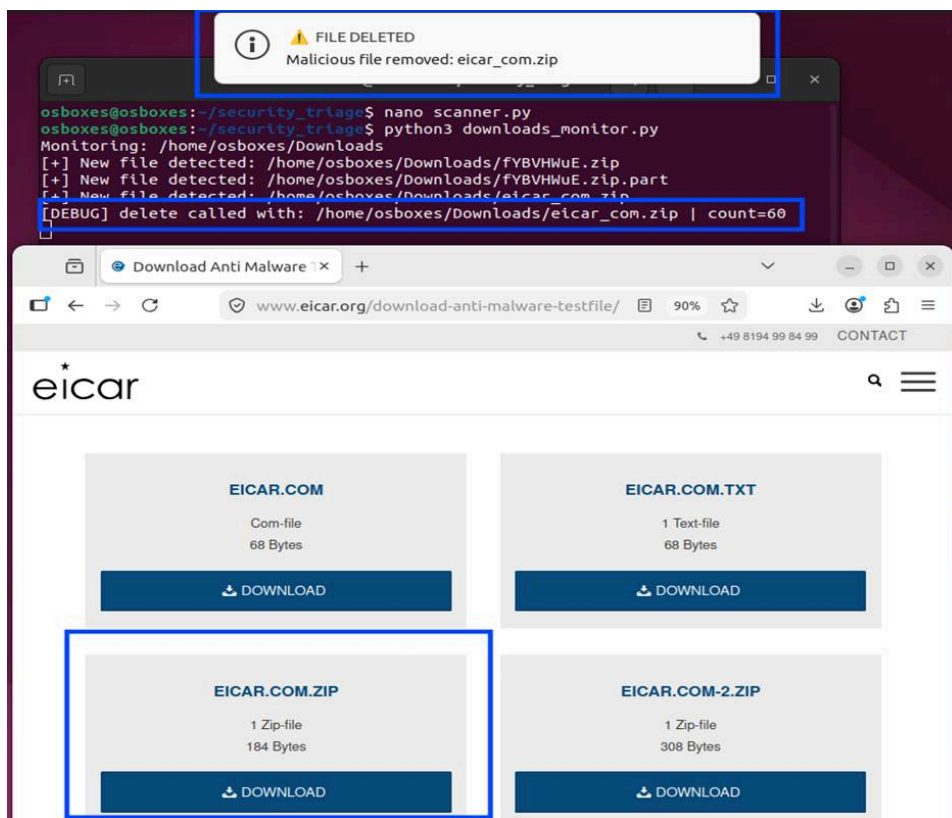
The popup should show:

```
FILE DELETED
Malicious File removed: eicar_com.zip
```

Press Ctrl C in Terminal 1 to stop the monitor once verified.

To confirm Virustotal flagged the hash and `safe_delete_file` executed.

Run `ls ~/Downloads/eicar.com.txt` in Terminal 2. It will confirm the file is gone- *No such file or directory*.



Conclusion

This lab brought together real-time file monitoring, Virustotal threat intelligence, and automated containment into a light EDR setup. By instantly hashing incoming downloads and deleting anything flagged as malicious, the system neutralizes threats on the spot without relying on manual checks. It turned out to be a solid, hands-on way to showcase automated incident response and endpoint defense in action.